

Sr-Nd-Os-S isotope and PGE geochemistry of the Xiarihamu magmatic sulfide deposit in the Qinghai–Tibet plateau, China

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Abstract The newly discovered Xiarihamu Ni-Cu deposit is located in the Eastern Kunlun orogenic belt in the northern part of the Qinghai–Tibet plateau, western China. It is the largest magmatic Ni-Cu sulfide deposit found thus far in an arc setting worldwide and ranks second in China in terms of total Ni resources. Fe-Ni-Cu sulfide mineralization occurs in a small ultramafic body that is part of a larger mafic-ultramafic complex formed by protracted Silurian–Early Devonian basaltic magmatism. The mineralized ultramafic body is composed predominantly of lherzolite and olivine websterite, with minor dunite, websterite and orthopyroxenite. Here we report new PGE (platinum group element) data and the results of a new, integrated Sr-Nd-Os-S isotope study. The initial concentrations of Rh and Pd in the parental magma are estimated to be 0.014 ppb and 0.24 ppb, respectively, which are more than one order of magnitude lower than those in undepleted mantle-derived magmas such as many continental picrites. The observed PGE depletions in the Xiarihamu parental magma are attributed to sulfide retention in the source mantle, because the degree of partial melting required to generate

the Xiarihamu primary magma was not high enough for a magma of that composition to dissolve all sulfides in the source. The $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratios and $\varepsilon_{\text{Nd}}(t)$ of the Xiarihamu host rocks range from 0.7062 to 0.7105 and from -1.97 to -5.74 , respectively, indicating 5–30 wt% crustal contamination in the Xiarihamu magma. These data also reveal that the source mantle for the Xiarihamu magma is isotopically (Sr-Nd) more enriched than that for the average Cenozoic arc basalt. The $\gamma_{\text{Os}}(t)$ and $\delta^{34}\text{S}$ values of sulfide ores from the Xiarihamu deposit range from 78 to 1393 and from 2 to 6‰, respectively. These values clearly indicate addition of crustal Os and S to the Xiarihamu parental magma. Metal tenors such as Ni and Rh are inversely correlated with $\gamma_{\text{Os}}(t)$ and $\delta^{34}\text{S}$ values. This indicates that mixing between different pulses of magma with different isotope compositions took place during magma emplacement at Xiarihamu, and that external S played a critical role in the formation of the Xiarihamu magmatic Ni-Cu sulfide deposit.

Introduction

The discovery of the Xiarihamu magmatic Fe-Ni-Cu sulfide deposit in 2011 (Li et al. 2012) represents the most significant breakthrough in global Ni exploration since the discovery of the Voisey's Bay Ni-Cu-Co deposit about three decades ago. It is also the largest magmatic Ni-Cu deposit ever discovered in an arc setting worldwide (Li et al. 2015) and ranks second in China in terms of total Ni resources, only after the Jinchuan deposit, which is the largest single sulfide Ni-Cu ore deposit in the world (Li and Ripley 2011). The Xiarihamu deposit is hosted in a small ultramafic body within an arc-type mafic-ultramafic complex in the East Kunlun orogenic belt, northern Tibet–Qinghai plateau, western China. The first scientific reports in English for this newly discovered deposit were given by Li

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et al. (2015) and Song et al. (2016). These authors provided important geochronological and petrological data, but no Sr–Nd–Os isotope or integrated PGE isotope data for the deposit. Here, we report these new types of data plus additional olivine analyses and S isotopes to complement the new isotope and PGE data. We use these data to address questions such as the controls on metal tenor variations, the cause of PGE depletion in the deposit, the degree of crustal contamination, the role of external sulfur in ore genesis, and the nature of the source mantle.

Geological background

The Xiarihamu Ni–Cu sulfide deposit is hosted in a mafic-ultramafic complex that occurs in the East Kunlun orogenic belt in the northern part of the Qinghai–Tibet plateau (Fig. 1a). The East Kunlun orogenic belt is the eastern segment of the E–W trending Kunlun Paleozoic–Mesozoic orogenic belt. The western segment of the Kunlun orogenic belt is commonly referred to as the West Kunlun orogenic belt in the literature (e.g., Xiao et al. 2005). These two segments have been displaced from each other by ~475 km since the middle Oligocene by the still-active sinistral strike-slip Altyn Tagh fault (Tapponnier et al. 2003). The East Kunlun orogenic belt merges with the Qinling orogenic belt in the east (Fig. 1a). To the north occur the Qaidam and Qilian microcontinental blocks (Fig. 1b), which are thought to be Gondwana-derived by some researchers (e.g., Metcalfe 2013). The suture between these two blocks is marked by a 441–457 Ma eclogite belt (Song et al. 2006; Xiong et al. 2012). Farther to the north, another Paleozoic tectonic suture, the North Qilian orogenic belt, occurs between the Qilian block and the Alxa block (Fig. 1b). The U–Pb ages of zircon crystals in eclogites in the North Qilian orogenic belt are ~464 Ma (Song et al. 2006), similar to the ages of zircon crystals in eclogites in the suture between the Qilian and Qaidam blocks. Several magmatic Ni–Cu occurrences with different ages occur in these two blocks (Fig. 1b). These include the 402 Ma Niubiziliang occurrence in the western part of the Qaidam block (Qian et al. 2015) and the 441–443 Ma Yaqu and Yulonggou occurrences in the southeastern part of the Qilian block (Zhang et al. 2014b). Farther to the north in the southern margin of the North China Craton occurs the Jinchuan magmatic sulfide Ni–Cu deposit that formed at ~831 Ma (Zhang et al. 2010).

The East Kunlun orogenic belt is further divided into the South and North Kunlun orogenic belts along the central Kunlun regional fault (Fig. 1b). The Xiarihamu Ni–Cu ore-bearing mafic-ultramafic complex is located at the northern rim of the East Kunlun orogenic belt. Abundant granitoids with zircon U–Pb ages varying from 391 to 432 Ma (Zhang et al. 2014a; Meng et al. 2015; Wang et al. 2013) are present in the North Kunlun orogenic belt (Fig. 1c). The granitoids intruded both Precambrian metamorphic rocks and Paleozoic volcanic-

sedimentary strata. Triassic sedimentary rocks are present in several small areas. Eclogites in the eastern part have a zircon U–Pb age of ~428 Ma (Meng et al. 2013). Several ophiolite complexes with zircon U–Pb ages varying from 467 to 518 Ma (Bian et al. 2004; Meng et al. 2015; Yang et al. 1996) are present within the North Kunlun orogenic belt (Fig. 1c).

In the Xiarihamu area, there are four mafic-ultramafic intrusions or complexes (Fig. 2a). To date, only the no. 1 mafic-ultramafic complex contains economically valuable Fe–Ni–Cu sulfide mineralization. The no. 3 body is entirely composed of serpentinites, whereas the no. 4 body is composed of partially altered gabbros and pyroxenites (Jiang et al. 2015). The no. 2 mafic-ultramafic complex is exposed in two small areas ~800 m apart (Fig. 2a). Minor Fe–Ni–Cu sulfide mineralization occurs in the pyroxenite body (Jiang et al. 2015).

Geology of the Xiarihamu Ni–Cu deposit

The Xiarihamu no. 1 igneous complex consists of an older gabbroic intrusion with a LA–ICP–MS zircon U–Pb age of 431.3 ± 2.1 Ma and a younger ultramafic body with a LA–ICP–MS zircon U–Pb age of 411.6 ± 2.4 Ma (Li et al. 2015). Jiang et al. (2015) reported a LA–ICP–MS zircon U–Pb age of 439.1 ± 3 Ma for a gabbroic sample from the complex. Song et al. (2016) reported a SHRIMP zircon U–Pb age of 405.5 ± 2.7 Ma for a gabbroic sample from the complex and a SHRIMP zircon U–Pb age of 406.1 ± 2.7 Ma for a websterite sample from the complex. The results show that the mafic rocks in the complex formed by two separate intrusive events that took place ~25 Ma apart from each other. The younger one is broadly coeval with the ultramafic body.

The area of the Xiarihamu no. 1 igneous complex exposed on the surface is ~0.6 km wide and ~1.6 km long (Fig. 2b). Overall, the Xiarihamu mafic-ultramafic complex gently plunges to the west, with its westernmost part occurring >500 m below enclosing Precambrian metamorphic rocks (Fig. 2c). The western part of the Xiarihamu mafic-ultramafic complex is predominantly composed of ultramafic rocks that occur continuously to form a wedge-shaped intrusive body, generally dipping 20°–30° to the west. The upper part of the ultramafic intrusion has been partially eroded by weathering. Its middle and lower parts are surrounded by older gabbroic and metamorphic rocks. The total thicknesses of the middle and lower parts of the ultramafic body vary from ~450 to <150 m. Small, irregular blocks of older gabbroic rocks are preserved in places within the ultramafic intrusion (Fig. 2c–e).

Fe–Ni–Cu sulfide mineralization mainly occurs as disseminated to net-textured zones within the ultramafic body and near the contact with the older gabbro (Fig. 2c–e). The sizes of different sulfide zones tend to increase westward and downward. A large sulfide-rich body occurs in the lower, western

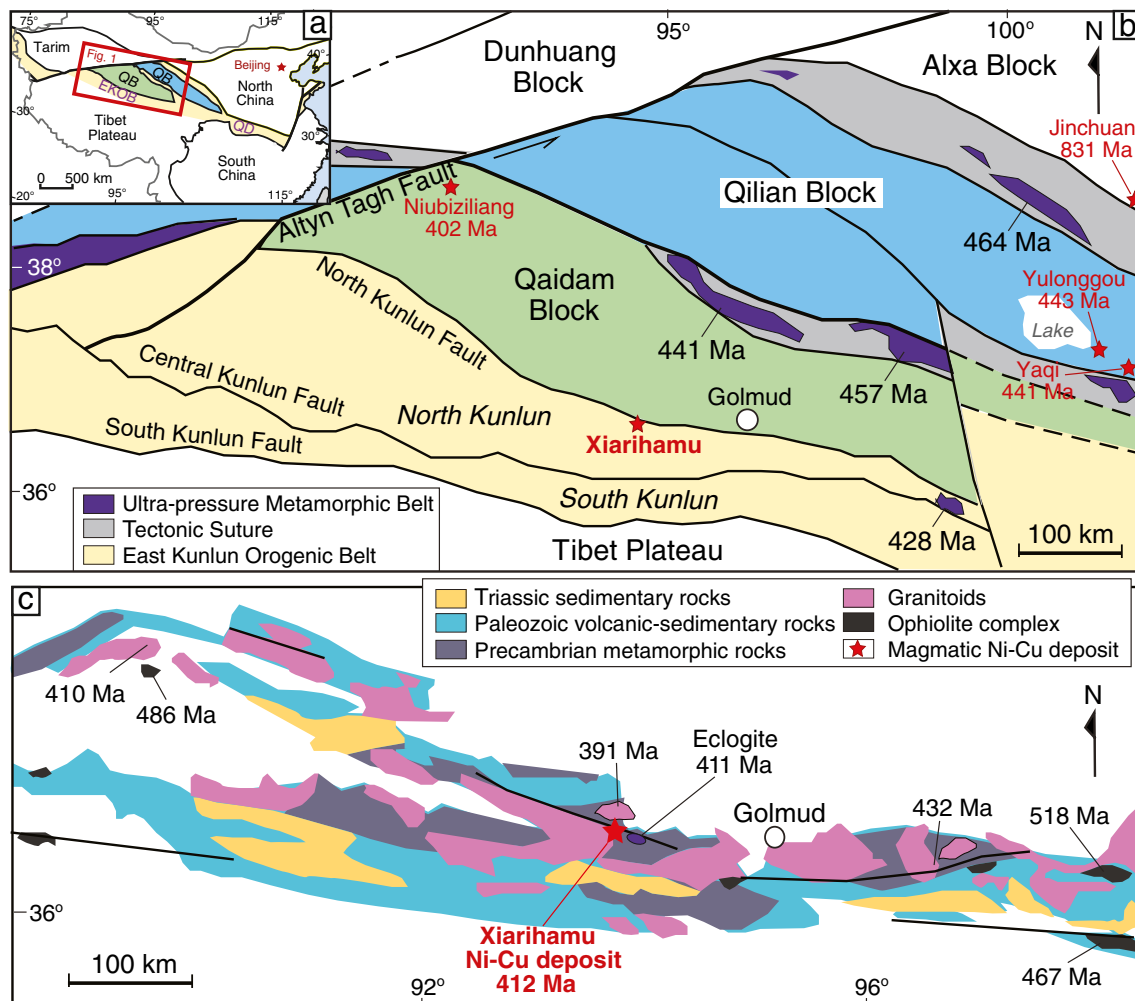


Fig. 1 Schematic map showing major continental blocks and orogenic belts in NW China (a), the tectonic units of the northern Qinghai–Tibet plateau (b), and simplified geology of the North Kunlun arc terrane (c), modified from Li et al. (2015). The zircon U–Pb isotope age data are from

Bian et al. (2004), Meng et al. (2013, 2015), Qi et al. (2014), Qian et al. (2015), Song et al. (2006), Wang et al. (2013), Xiong et al. (2012), Zhang et al. (2010), Zhang et al. (2014b), Zhang et al. (2014a), and Li et al. (2015)

part of the ultramafic body west of drill core ZK507 (Fig. 2c). In the eastern part of the ultramafic body, the sulfide zones are smaller and become isolated from each other (Fig. 2d, e).

Lherzolite, olivine websterite, and websterite are the major host rocks of the Xiarhamu deposit. They are all olivine or pyroxene cumulates. The ratio of orthopyroxene to clinopyroxene in these rocks is about 4 to 1 (Li et al. 2015). Plagioclase is very rare generally but is locally present in abundances up to ~8 % in the interstitial spaces of olivine cumulates. Small Cr-spinel inclusions are present in some olivine crystals. Some clinopyroxene crystals occur as large oikocrysts enclosing smaller olivine and orthopyroxene grains. Both clinopyroxene and orthopyroxene commonly occur as cumulus phases in websterite. Hornblende, biotite, and Fe–Ti oxides fill the interstitial spaces. Variable degrees of alteration are present in the rocks. Olivine is partially altered to serpentine; orthopyroxene is variably replaced by talc;

clinopyroxene is partially altered to actinolite; and plagioclase is partially replaced by clinozoisite and albite.

Sample descriptions

The samples used in this study were collected from drill cores in the western part of the Xiarhamu ultramafic intrusion where the most important Fe–Ni–Cu sulfide mineralization occurs (Fig. 2b–e). Forty-five samples with different abundances of Fe–Ni–Cu sulfide mineralization were used in this study. Eleven samples were collected from the upper, eastern part of the deposit where the host rocks are mainly pyroxene cumulates. Thirty-four samples were collected from the lower, western part of the deposit where the host rocks are mainly olivine cumulates.

Our sample collection covers all types of Fe–Ni–Cu sulfide mineralization, including weakly disseminated, disseminated,

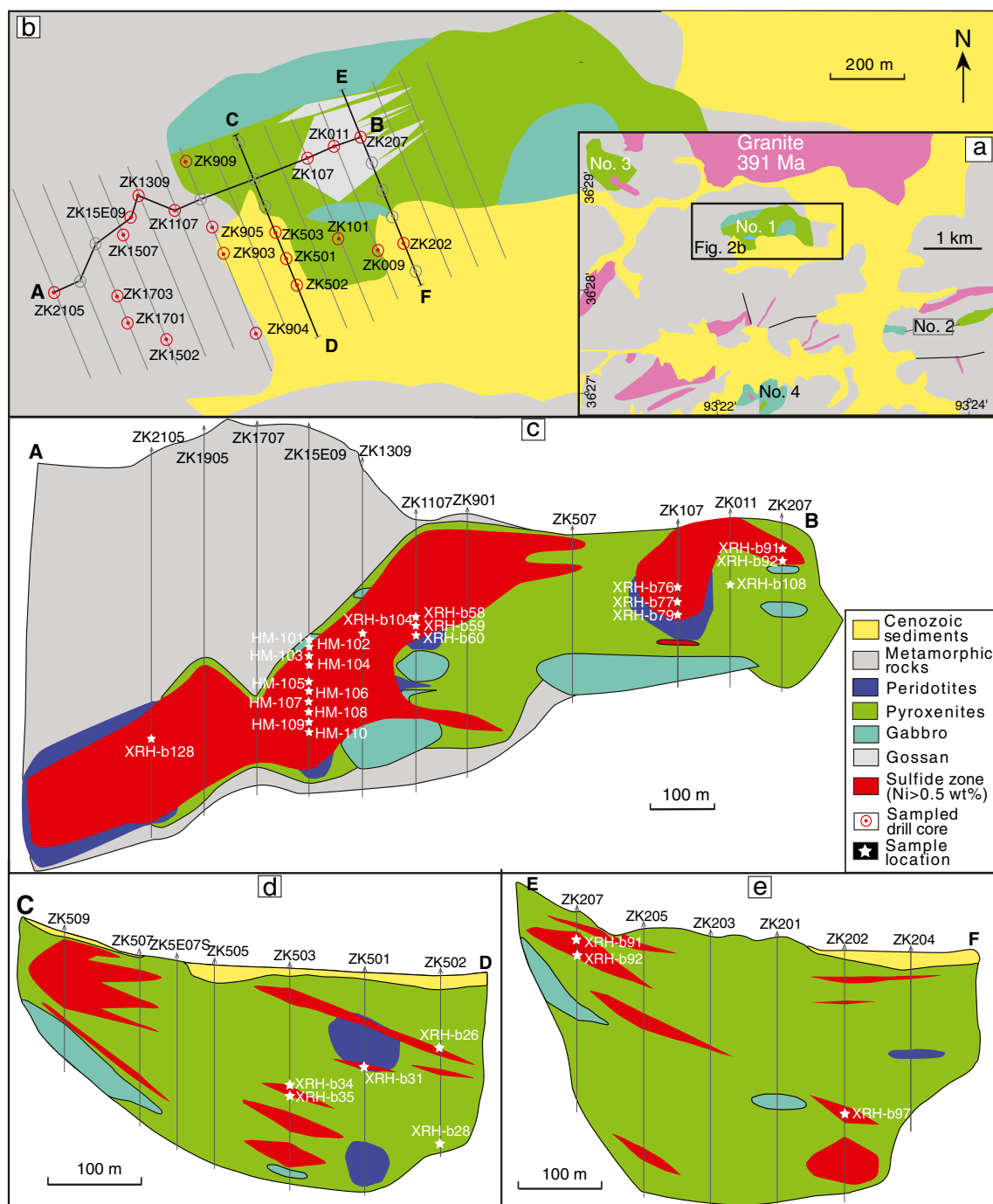


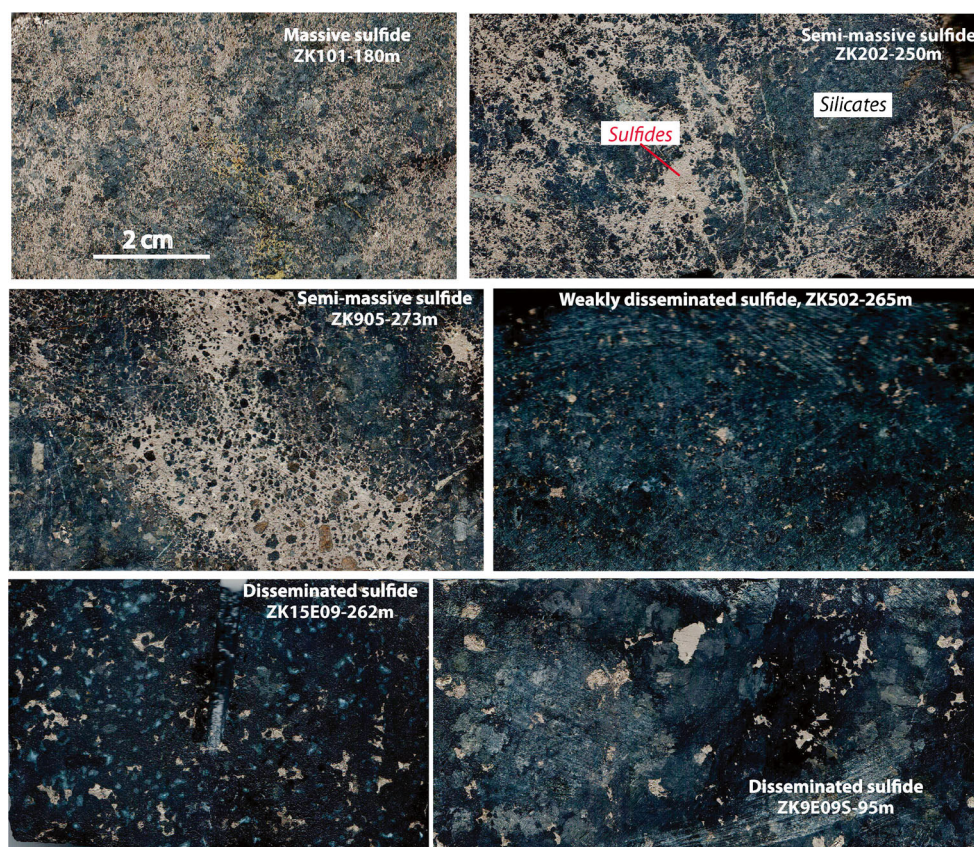
Fig. 2 Simplified geological map of the Xiarihamu area (a), and map and cross sections of the Xiarihamu magmatic sulfide deposit (b–e), modified from Li et al. (2015). The zircon U–Pb isotope ages are from Wang et al. (2013) and Li et al. (2015)

net-textured, semi-massive, massive and patchy sulfides, and sulfide veins. The representative textures are shown in Fig. 3. The sulfide assemblages are composed of pyrrhotite, pentlandite, and chalcopyrite. Minor amount of magmatic magnetite occurs as dissemination within the sulfide assemblages. Chlorite and secondary magnetite occur in the microfractures of base-metal sulfides. The rocks are more altered in areas affected by faulting and deformation and only partially altered in areas away from faulting and deformation.

Analytical methods

Olivine compositions were determined by wavelength-dispersive X-ray analysis using a CAMECA SX50 electron microprobe at Indiana University. The analytical conditions for major and minor elements were 15 kV, 20-nA beam current, 1-μm beam size, and peak-counting time of 20 s. Nickel and Ca in olivine were analyzed using a beam current of 100 nA and a peak-counting time of 50 s. The detection limits

Fig. 3 Photographs of selected sulfide ore samples from the Xiarihamu Ni-Cu deposit



for Ni and Ca under such conditions are 100 and 50 ppm, respectively. Analytical uncertainty is within $\pm 2\%$ of accepted values, based on the results from the San Carlos olivine standard (USNM 1113122/444) analyzed together with our samples.

Whole-rock Ni and Cu were analyzed by inductively coupled plasma-mass spectrometry (ICP-MS) using a Thermo Elemental X-series instrument in the National Research Center of Geoanalysis in Beijing, China. The concentrations of sulfur in whole rocks were determined by combustion-iodometric titration using a LECO CS230 instrument at the Institute of Geology and Mineral Resources, Xi'an, China. The concentrations of PGEs in sulfide-bearing samples were determined by the combination of NiS fire-assay preconcentration and Te coprecipitation, followed by ICP-MS analysis in the National Research Center of Geoanalysis in Beijing, China. The analytical uncertainty is within $\pm 10\%$ of accepted values, based on the results from the international standards (UMT-1 and WPR-1) analyzed together with our samples.

Sulfur isotope analysis was carried out at Indiana University using the continuous-flow method described in Studley et al. (2002). Fe-Ni-Cu sulfide minerals (pentlandite, chalcopyrite, and pyrrhotite) were drilled from polished sections using a 0.75-mm carbide bit. Between 0.1 and 0.2 mg of sulfide powder was placed in tin cups with

approximately 1.5 to 2 mg vanadium pentoxide (V_2O_5). Samples were prepared in an elemental analyzer by flash combustion at 1800 °C with a reactor column temperature of 1010 °C. Measurements of produced SO_2 were made using a Finnigan Delta V Stable isotope ratio mass spectrometer, with results reported in per mil delta notation relative to Vienna-Canyon Diablo Troilite (VCDT). Analytical uncertainty was less than $\pm 0.05\%$, and sample reproducibility was within $\pm 0.2\%$. Sulfide standards used were IAEA-S1, IAEA-S2, and IAEA-S3, with values of -0.3 , 21.7 , and -31.3% , respectively, on the SO_2 scale.

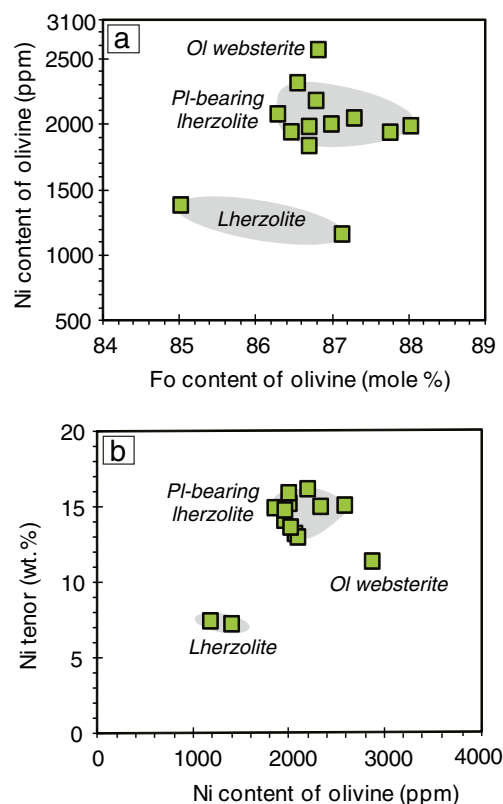
Whole-rock Rb-Sr and Sm-Nd isotopes were determined using a Nu Plasma HR MC-ICP-MS in the State Key Laboratory of Continental Dynamics, Northwest University, Xi'an, China. Procedural blanks were 40 pg for Rb, 300 pg for Sr, 20 pg for Sm, and 60 pg for Nd. Mass fractionation corrections for Sr and Nd isotope ratios were based on the values of $^{86}Sr/^{88}Sr = 0.1194$ and $^{146}Nd/^{144}Nd = 0.7219$. The measured values for NBS-987 Sr standard and the JNdi-1 Nd standard were $^{87}Sr/^{86}Sr = 0.710245 \pm 16$ (2σ , $n = 8$) and $^{143}Nd/^{144}Nd = 0.512117 \pm 10$ (2σ , $n = 10$), respectively. The USGS reference material BCR-2 was used to monitor the accuracy. The results for this reference sample were 46.62 ppm Rb, 367.49 ppm Sr, $^{87}Sr/^{86}Sr = 0.704969 \pm 3$ (2σ), 6.473 ppm Sm, 27.62 ppm Nd, and $^{143}Nd/^{144}Nd = 0.512674 \pm 10$ (2σ).

Table 1 Average composition of olivine in sulfide-bearing samples from the Xiarihamu Ni-Cu deposit

Sample	Drill core	Depth (m)	Rock type	n	SiO ₂	MgO	FeO	MnO	CaO	NiO	Total	Si	Mg	Fe	Mn	Ca	Ni	Total	Fo	Ni
XRH-b2	ZK903	140	Lherzolite	10	39.77	46.79	14.49	0.19	0.01	0.18	101.42	0.99	1.70	0.30	0.004	0.000	0.004	3.00	85.04	1394
XRH-b48	ZK1703	301	Lherzolite	9	40.72	46.99	12.22	0.18	0.00	0.15	100.26	1.01	1.73	0.25	0.004	0.000	0.003	2.99	87.11	1172
XRH-b58	ZK1107	135	Ol websterite	8	40.66	46.83	12.57	0.14	0.01	0.33	100.55	1.00	1.72	0.26	0.003	0.000	0.007	3.00	86.80	2578
HM-101	ZK15E09	259	Pl-bearing lherzolite	7	40.21	47.01	12.08	0.17	0.02	0.26	99.75	1.00	1.74	0.25	0.003	0.000	0.005	3.00	87.27	2057
HM-102	ZK15E09	262	Pl-bearing lherzolite	10	40.45	47.75	11.75	0.15	0.01	0.25	100.36	1.00	1.76	0.24	0.003	0.000	0.005	3.00	87.74	1948
HM-104	ZK15E09	292	Pl-bearing lherzolite	9	40.48	46.54	13.06	0.16	0.02	0.27	100.53	1.00	1.72	0.27	0.003	0.001	0.005	3.00	86.27	2088
HM-105	ZK15E09	339	Pl-bearing lherzolite	9	40.08	46.69	12.53	0.17	0.01	0.28	99.75	1.00	1.73	0.26	0.004	0.000	0.006	3.00	86.77	2190
HM-106	ZK15E09	343	Pl-bearing lherzolite	9	40.74	47.76	11.44	0.17	0.01	0.25	100.36	1.00	1.75	0.24	0.004	0.000	0.005	3.00	88.01	1997
HM-107	ZK15E09	369	Pl-bearing lherzolite	10	40.45	46.75	12.70	0.17	0.06	0.23	100.36	1.00	1.72	0.26	0.004	0.002	0.005	3.00	86.68	1846
HM-108	ZK15E09	375	Pl-bearing lherzolite	8	40.81	47.08	12.43	0.20	0.04	0.26	100.81	1.00	1.73	0.26	0.004	0.001	0.005	3.00	86.97	2011
HM-109	ZK15E09	381	Pl-bearing lherzolite	8	40.13	46.99	12.71	0.18	0.02	0.25	100.28	1.00	1.74	0.26	0.004	0.000	0.005	3.00	86.68	1992
HM-110	ZK15E09	390	Pl-bearing lherzolite	7	40.45	46.57	12.73	0.21	0.01	0.30	100.26	1.00	1.72	0.26	0.004	0.000	0.006	3.00	86.53	2325

Oxide in wt%, Fo in mol%, Ni in ppm

Re and Os isotopes were determined using a PE ELAN DRC-e ICP-MS in the National Research Center of Geoanalysis in Beijing, China. The concentrations of Re and Os were determined by isotope dilution. The sample preparation and analytical procedures are similar to those given in Du et al. (2004). About 2.5 g powdered sample was placed in a Carius tube together with ^{185}Re and ^{190}Os mixed spikes and digested by reverse aqua regia with H_2O_2 (Shirey and Walker 1995). Os was separated as OsO_4 by distillation at 105–110 °C and was trapped by Milli-Q water. Re was extracted from the residue by acetone in 5 M NaOH solution. Re concentrations were measured by a PQ Excell ICP-MS. The intensities of ^{190}Os and ^{185}Re were used to correct the interference between Os and Re. All measured ratios were corrected for oxygen isotopic compositions (Sun et al. 1997), and mass fractionation and Os concentrations and isotopic compositions were calculated iteratively. The total procedural blanks were about 2.9 pg for Re and 1 to 0.2 pg for Os. The internal standard JCBY (the Jinchuan sulfide ore) was used to monitor the accuracy of the measurements. The Re and Os contents and $^{187}\text{Os}/^{188}\text{Os}$ ratio of standard JCBY determined during the course of this work are 38.65 ± 0.12 ppb, 15.65 ± 0.05 ppb, and 0.3361 ± 0.0017 , respectively, which are similar to the certified values.

**Fig. 4** Plots of Ni vs Fo in olivine (a) and Ni tenor vs Ni in olivine (b) for the Xiarihamu deposit

Analytical results

Olivine compositions

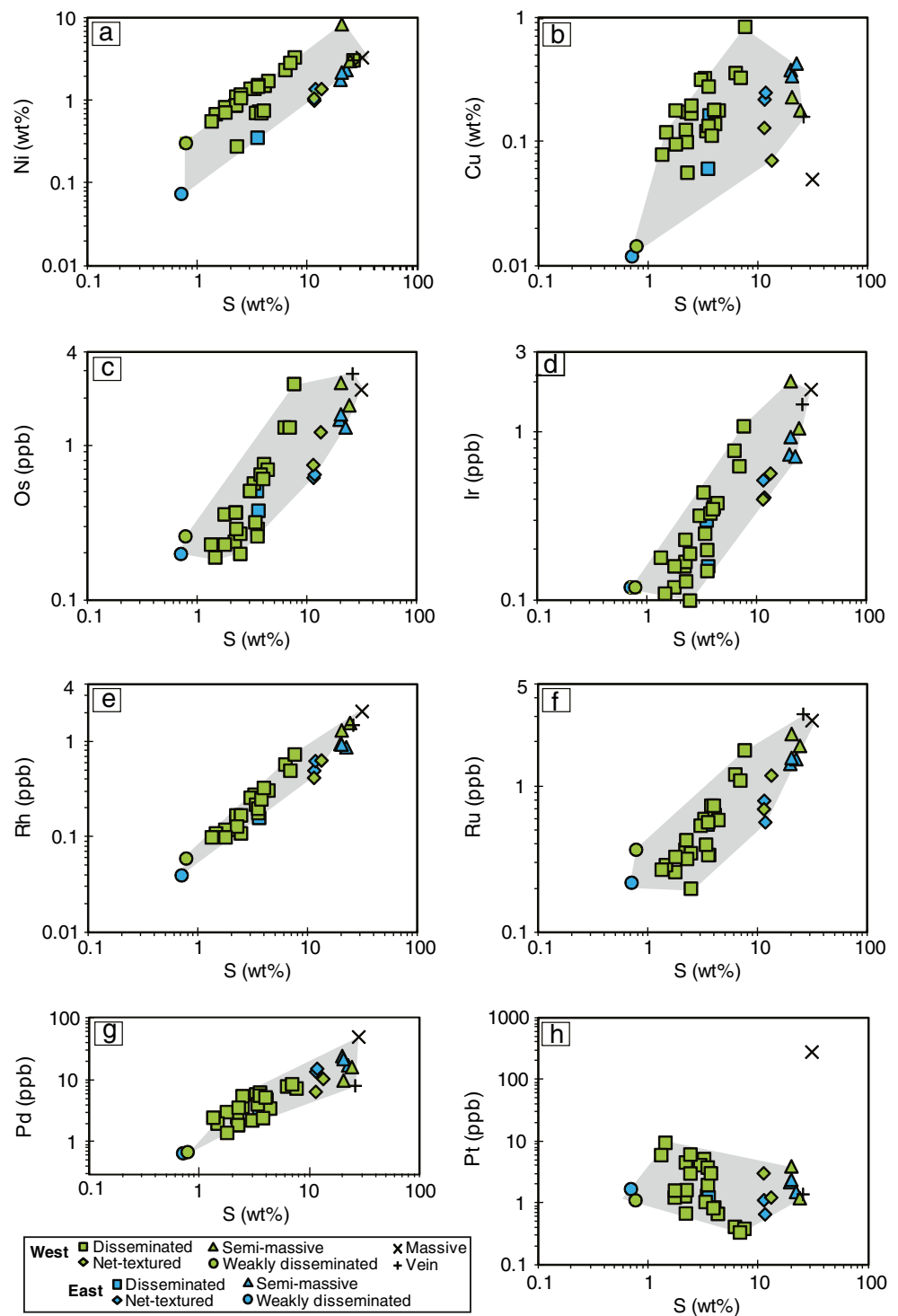
Average compositions of olivine in sulfide-bearing samples from the Xiarihamu Ni-Cu deposit are listed in Table 1. The samples used in this study are all from the lower, western part of the deposit. The rock types of the samples are plagioclase-

bearing lherzolite, lherzolite, and olivine websterite. The Fo contents of olivine in the samples vary from 85 to 88 mol%. Olivine in two lherzolite samples show significant Ni depletion (<1400 ppm Ni) as compared to the plagioclase-bearing lherzolite samples (1846–2325 ppm). Olivine in an olivine websterite sample has an intermediate Fo content (86.8 mol%) but the highest Ni content (2578 ppm). Olivine in lherzolite samples shows a negative correlation between Ni

Table 2 Concentrations of S, Ni, and Cu (in wt%) and PGE (in ppb) in sulfide-bearing samples from the Xiarihamu Ni-Cu deposit

Sample no.	Drill core	Depth (m)	Orebody	Host rock	Sulfide texture	S	Ni	Cu	Os	Ir	Ru	Rh	Pt	Pd
XRH-b61	ZK009	115	East	Websterite	Disseminated	3.45	0.37	0.06	0.51	0.30	0.55	0.22	1.34	3.67
XRH-b91	ZK207	41	East	Websterite	Disseminated	3.56	0.76	0.16	0.38	0.16	0.34	0.16	1.25	4.24
XRH-b97	ZK202	235	East	Websterite	Net-textured	11.32	1.03	0.22	0.62	0.52	0.80	0.50	1.13	14.10
XRH-b92	ZK207	57	East	Websterite	Net-textured	11.58	1.42	0.25	0.65	0.41	0.57	0.63	0.67	15.80
XRH-b117	ZK101	182	East	Websterite	Semi-massive	19.56	1.84	0.38	1.47	0.74	1.43	0.95	2.22	25.40
XRH-b76	ZK107	108	East	Websterite	Semi-massive	22.13	2.42	0.43	1.31	0.72	1.54	0.88	1.54	17.90
XRH-b77	ZK107	122	East	Websterite	Semi-massive	20.07	2.26	0.34	1.59	0.94	1.57	0.94	2.42	22.70
XRH-b79	ZK107	150	East	Websterite	Weakly disseminated	0.70	0.08	0.01	0.20	0.12	0.22	0.04	1.72	0.68
XRH-b108	ZK011	130	East	Websterite	Patchy				1.43	1.19	1.81	1.16	2.39	25.50
XRH-b58	ZK1107	135	West	Ol websterite	Disseminated	3.23	1.43	0.33	0.57	0.44	0.60	0.28	5.25	6.08
XRH-b60	ZK1107	152	West	Websterite	Disseminated	2.98	1.44	0.32	0.51	0.32	0.54	0.26	4.18	2.30
XRH-b104	ZK1309	231	West	Lherzolite	Disseminated	2.21	1.15	0.18	0.37	0.23	0.34	0.15	4.62	1.94
XRH-b43	ZK1507	444	West	Lherzolite	Disseminated	2.17	0.90	0.13	0.24	0.16	0.37	0.12	1.31	2.60
HM-101	ZK15E09	259	West	Pl-bearing lherzolite	Disseminated	4.00	1.56	0.14	0.76	0.36	0.64	0.31	0.86	5.01
HM-102	ZK15E09	262	West	Pl-bearing lherzolite	Disseminated	4.32	1.78	0.18	0.70	0.38	0.59	0.31	0.68	3.59
HM-103	ZK15E09	276	West	Pl-bearing lherzolite	Disseminated	1.75	0.86	0.18	0.36	0.12	0.26	0.12	1.25	3.14
HM-104	ZK15E09	292	West	Pl-bearing lherzolite	Disseminated	2.21	0.91	0.10	0.37	0.17	0.43	0.17	0.69	1.94
HM-105	ZK15E09	339	West	Pl-bearing lherzolite	Disseminated	2.43	1.22	0.17	0.20	0.10	0.20	0.11	3.06	3.84
HM-106	ZK15E09	343	West	Pl-bearing lherzolite	Disseminated	3.50	1.59	0.28	0.29	0.15	0.34	0.18	2.00	6.50
HM-107	ZK15E09	369	West	Pl-bearing lherzolite	Disseminated	1.44	0.70	0.12	0.19	0.11	0.29	0.11	9.63	2.05
HM-108	ZK15E09	375	West	Pl-bearing lherzolite	Disseminated	6.17	2.42	0.36	1.31	0.78	1.21	0.58	0.42	8.17
HM-109	ZK15E09	381	West	Pl-bearing lherzolite	Disseminated	7.52	3.45	0.85	2.50	1.09	1.77	0.74	0.39	7.57
HM-110	ZK15E09	390	West	Pl-bearing lherzolite	Disseminated	6.85	2.96	0.33	1.31	0.63	1.10	0.50	0.34	8.88
XRH-b56	ZK1701	298	West	Lherzolite	Disseminated	1.32	0.58	0.08	0.23	0.18	0.27	0.10	6.09	2.55
XRH-b46	ZK1703	244	West	Lherzolite	Disseminated	2.43	1.11	0.20	0.27	0.19	0.35	0.17	6.21	5.68
XRH-b48	ZK1703	301	West	Lherzolite	Disseminated	3.34	0.73	0.12	0.32	0.25	0.40	0.22	1.06	4.20
XRH-b128	ZK2105	443	West	Dunite	Disseminated	3.49	1.52	0.14	0.26	0.20	0.57	0.20	3.83	5.81
XRH-b31	ZK501	112	West	Orthopyroxenite	Disseminated	2.24	0.29	0.06	0.29	0.13	0.32	0.13	1.65	3.71
XRH-b26	ZK502	106	West	Lherzolite	Disseminated	1.76	0.74	0.10	0.23	0.16	0.33	0.10	1.62	1.45
XRH-b2	ZK903	140	West	Lherzolite	Disseminated	3.73	0.72	0.11	0.65	0.33	0.74	0.25	3.08	2.51
XRH-b10	ZK905	152	West	Websterite	Disseminated	3.93	0.78	0.18	0.61	0.35	0.74	0.33	0.84	5.43
XRH-b35	ZK503	163	West	Websterite	Massive	30.87	3.42	0.05	2.29	1.80	2.83	2.09	283	57.2
XRH-b34	ZK503	130	West	Websterite	Net-textured	13.16	1.42	0.07	1.22	0.57	1.19	0.64	1.25	10.80
XRH-b18	ZK909	175	West	Websterite	Net-textured	11.25	1.09	0.13	0.75	0.40	0.70	0.42	3.08	6.68
XRH-b59	ZK1107	140	West	Websterite	Semi-massive	20.10	8.70	0.23	2.54	2.02	2.29	1.32	3.99	10.20
XRH-b11	ZK905	273	West	Websterite	Semi-massive	23.94	3.12	0.18	1.82	1.06	1.90	1.58	1.22	16.80
XRH-b4	ZK903	214	West	Websterite	Vein	25.64	3.17	0.16	2.91	1.47	3.13	1.50	1.41	8.36
XRH-b28	ZK502	265	West	Lherzolite	Weakly disseminated	0.77	0.31	0.01	0.26	0.12	0.37	0.06	1.13	0.71

Fig. 5 Plots of chalcophile element concentrations vs S content in whole rocks from the Xiarihamu deposit



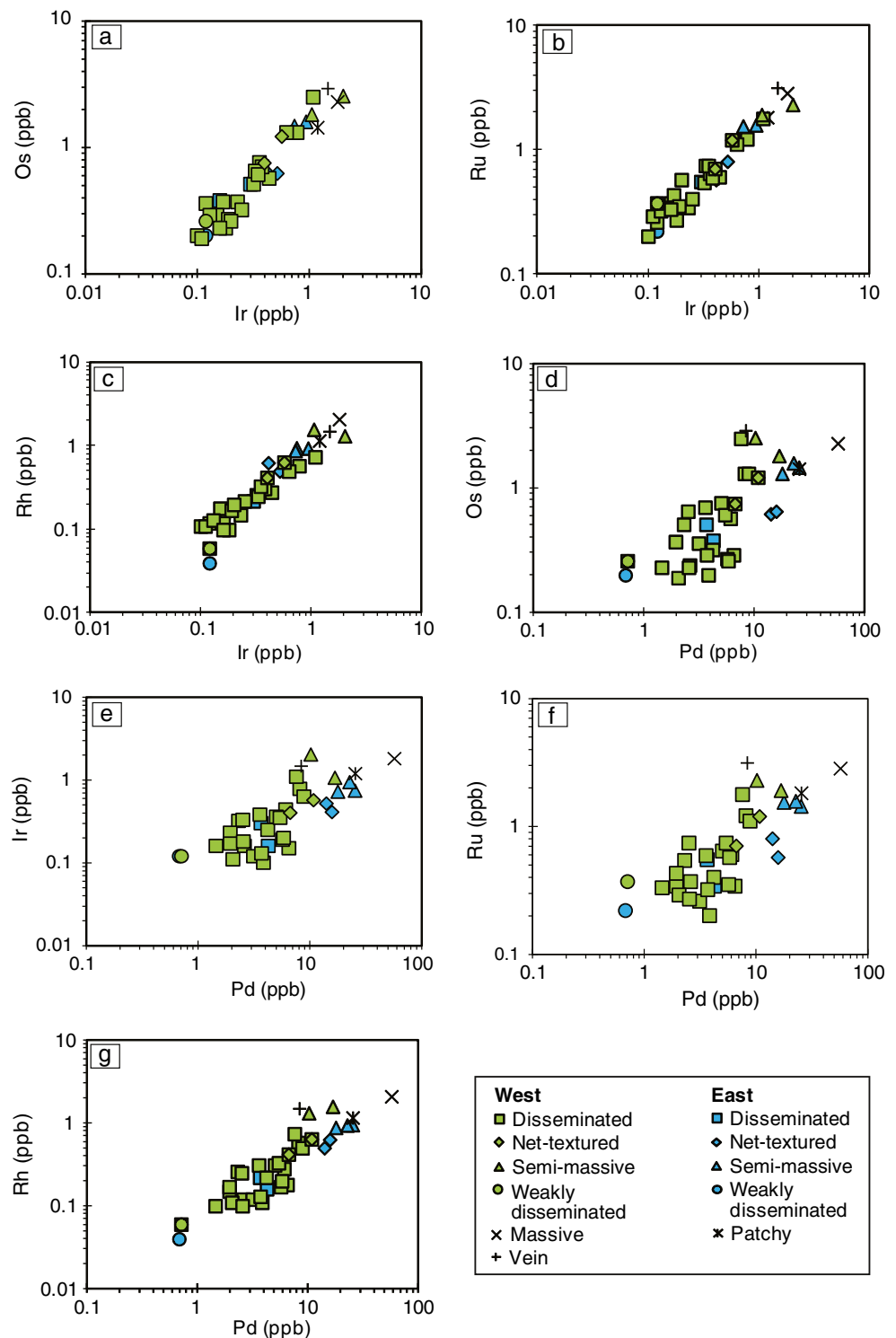
and Fo contents, whereas olivine in plagioclase-bearing hercynite samples does not exhibit such correlation (Fig. 4).

Ni, Cu, and PGE concentrations

The concentrations of Cu, Ni, PGEs, and S in sulfide-bearing samples from the Xiarihamu deposit are listed in

Table 2. In the following presentation, we use the Ni values after correction for the contribution from olivine in the samples. The correction was done using the observed olivine contents in the samples and the average Ni content of olivine from each sample. At a given S content, samples from the upper, eastern part of the deposit tend to have lower Ni contents (Fig. 5a). With the exception of the only

Fig. 6 Relationships between chalcophile elements in whole rocks from the Xiarihamu deposit



massive sulfide sample, we have analyzed, all of the other samples from the deposit show positive correlations between S and the selected chalcophile elements except Pt (Fig. 5). Among all IPGEs (Ir, Os, Ru, and Rh), Rh has the best correlation with S. Excluding the massive sulfide sample, the rest of the samples show a very poor negative correlation between Pt and S (Fig. 5h).

Excellent positive correlations are present between IPGEs (Fig. 6a–c). Positive correlations are also present between Pd and individual IPGEs (Fig. 6e–g), albeit less perfect than those between IPGEs. Among all IPGEs, Rh has the best correlation with Pd (Fig. 6g).

The primitive mantle-normalized patterns of Cu, Ni, and PGEs in bulk sulfide (recalculated to 100 % sulfide

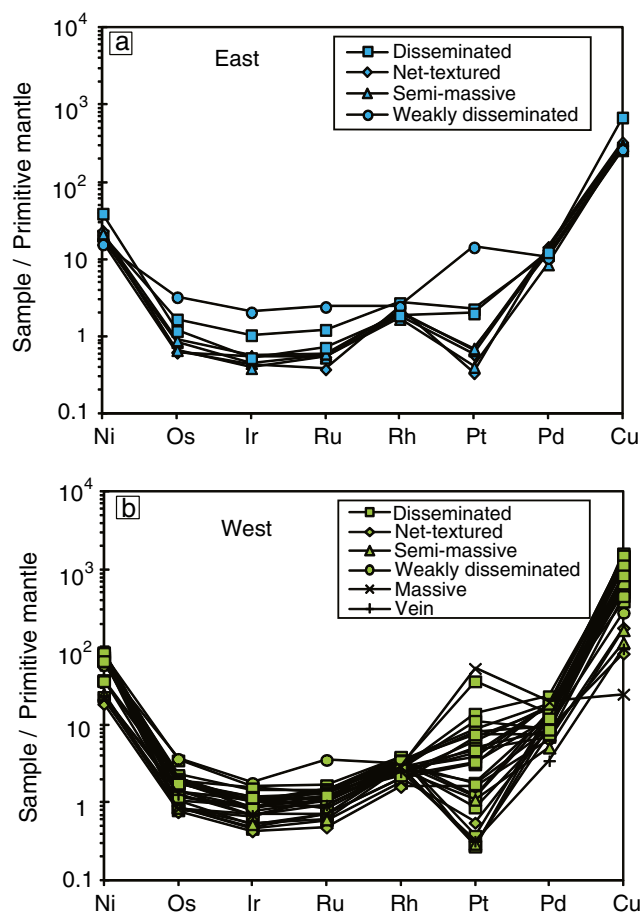


Fig. 7 Primitive mantle-normalized Ni-PGE-Cu patterns of sulfide-bearing samples from the Xiarihamu deposit. The normalization values are from Palme and O'Neill (2014)

(i.e., metal tenors) are illustrated in Fig. 7. The tenor calculation was done using the equation of Barnes and Lightfoot (2005), which is based on the assumption that the compositions of pentlandite and pyrrhotite are $\text{Fe}_{4.5}\text{Ni}_{4.5}\text{S}_8$ and Fe_7S_8 , respectively. The average compositions of these minerals in the Xiarihamu deposits are very similar to the assumed values (i.e., <5 %). All of the samples show significant PGE depletion relative to Cu and Ni. Most of the samples show a negative or positive Pt anomaly, which explains the lack of correlation between Pt and other chalcophile elements including Pd described above. The ranges of PGE tenors in the eastern and western parts of the deposit are similar, but some samples from the western part have higher Ni tenors.

For the two types of lherzolite from the western part of the deposit, higher Ni tenors are coupled with higher Ni contents in olivine (Fig. 4b). An olivine websterite sample from this part of the deposit has intermediate Ni tenor but the highest Ni content in olivine among all of the samples that have been analyzed for both olivine and whole rock compositions (Fig. 4b).

Table 3 Rb-Sr-Sm-Nd data for the Xiarihamu Ni-Cu deposit

Sample no.	Rock type	Drill core	Depth (m)	Rb (ppm)	Sr (ppm)	$^{87}\text{Rb}/^{86}\text{Sr}$	$^{87}\text{Sr}/^{86}\text{Sr}$	2σ	Sm (ppm)	Nd (ppm)	$^{147}\text{Sm}/^{144}\text{Nd}$	$^{143}\text{Nd}/^{144}\text{Nd}$	2σ	$(^{87}\text{Sr}/^{86}\text{Sr})_i$	$\epsilon_{\text{Nd}(0)}$
XRH-b48	Lherzolite	ZK1703	301	2.32	25.7	0.261189	0.708031	0.000009	0.37	1.22	0.183343	0.512488	0.000013	0.706500	-2.22
XRH-b28	Lherzolite	ZK1502	265	23.7	16.3	4.218055	0.735188	0.000019	0.53	1.77	0.181012	0.512302	0.000014	0.710463	-5.74
XRH-b2	Lherzolite	ZK901	140	1.48	10.1	0.424111	0.711328	0.000008	0.38	1.12	0.205111	0.512491	0.000015	0.708842	-3.32
XRH-b15	Lherzolite	ZK904	215	3.36	47	0.206916	0.711618	0.000007	0.4	1.27	0.190407	0.512520	0.000014	0.710405	-1.97
XRH-b1	Orthopyroxenite	ZK901	109	6.52	48.1	0.392319	0.711274	0.000010	1.36	4.12	0.199555	0.512466	0.000009	0.708974	-3.51
XRH-b1-dp	Orthopyroxenite	ZK901	109	6.52	48.1	0.392320	0.711294	0.000008	1.36	4.12	0.199554	0.512442	0.000007	0.708995	-3.97
XRH-b31	Orthopyroxenite	ZK501	112	6.09	24	0.734379	0.710730	0.000005	0.5	1.67	0.180997	0.512433	0.000009	0.706426	-3.17
XRH-b8	Gabbro	ZK905	112	117	494	0.685638	0.713602	0.000009	6.76	24.9	0.164115	0.512257	0.000004	0.709390	-5.65

ϵ_{Nd} is the derivation in parts per 10,000 of the initial ratio from that of a chondritic reservoir at the crystallization age of 431.3 Ma for gabbro

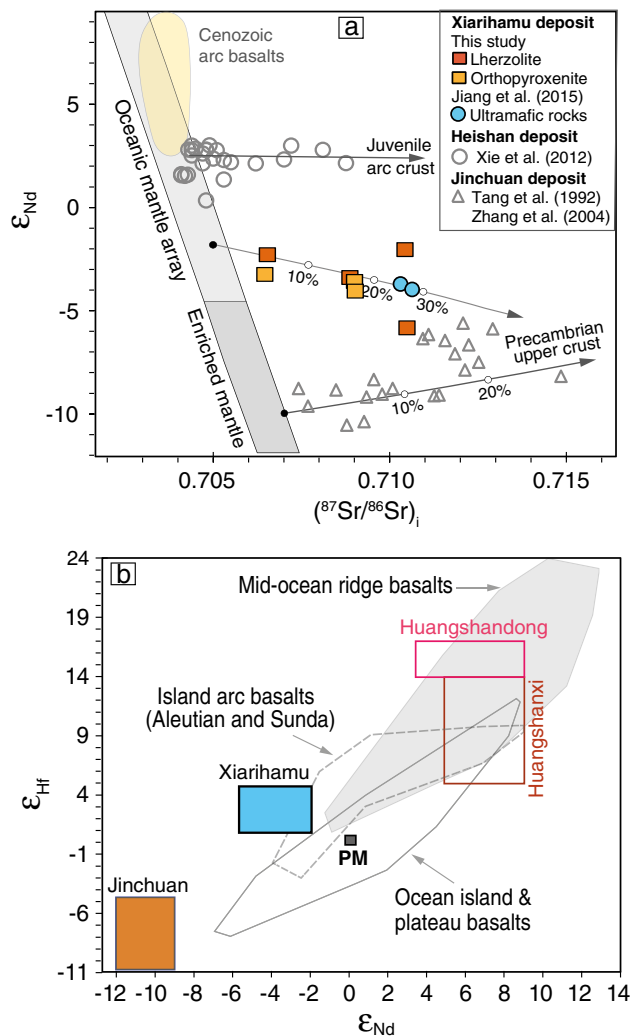


Fig. 8 Plots of whole rock $\varepsilon_{\text{Nd}}(t)$ vs $(^{87}\text{Sr}/^{86}\text{Sr})_i$ (a) and zircon $\varepsilon_{\text{Hf}}(t)$ vs whole-rock $\varepsilon_{\text{Nd}}(t)$ (b) for the Xiarihamu deposit. The oceanic mantle array is from Zindler and Hart (1986). Values used in mixing calculations: upper crust, 27 ppm Nd, 320 ppm Sr, $\varepsilon_{\text{Nd}}(t) = -6$, $(^{87}\text{Sr}/^{86}\text{Sr})_i = 0.715$; mantle-derived magma, 9.8 ppm Nd, 97 ppm Sr, $\varepsilon_{\text{Nd}}(t) = -1.8$, $(^{87}\text{Sr}/^{86}\text{Sr})_i = 0.705$. Zircon ε_{Hf} data sources: Xiarihamu (Li et al. 2015), Jinchuan (Tang et al. 2014), Huangshandong (Sun et al. 2013), Huangshanxi (Su et al. 2011). Nd-Hf isotopes of basalts: Aleutian and Sunda arcs (Handley et al. 2011; Yogodzinski et al. 2010), midocean ridge and ocean island-plateau (Salters et al. 2011). The Sr-Nd isotopes of Cenozoic arc basalts worldwide are from a public database (<http://www.petdb.org>)

Rb-Sr and Sm-Nd isotopes

The Rb-Sr and Sm-Nd isotope data for the Xiarihamu deposit are given in Table 3. The initial isotopic ratios for the ultramafic rocks and associated gabbro were calculated using the zircon U-Pb ages of 411.6 and 431.3 Ma for these two different intrusive phases, respectively (Li et al. 2015). The $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratios of the ultramafic rocks from the Xiarihamu deposit range from 0.706426 to 0.710463. The older gabbro has a

$(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratio of 0.709390. The $\varepsilon_{\text{Nd}}(t)$ values of the ultramafic rocks and the older gabbro are from -1.97 to -5.74 and -5.65 , respectively.

A comparison with the Heishan magmatic Ni-Cu deposit associated with basaltic magmatism in an arc setting in western China (Xie et al. 2014), and the Jinchuan magmatic Ni-Cu deposit associated with intra-plate high-Mg basaltic magmatism in the North China Craton (Li and Ripley 2011) is illustrated in Fig. 8a. The $(^{87}\text{Sr}/^{86}\text{Sr})_i$ ratios of the Xiarihamu intrusion are similar to those of the Heishan deposit but slightly lower than those of the Jinchuan deposit. The $\varepsilon_{\text{Nd}}(t)$ values of the Xiarihamu deposit are significantly lower than those of the Heishan deposit but much higher than those of the Jinchuan deposit. The intercepted values of the observed Sr-Nd isotope arrays with the mantle array (Fig. 8a) indicate that the source mantle of the Xiarihamu deposit is isotopically more enriched than that of the Heishan deposit but not as enriched as that of the Jinchuan deposit.

The combination of co-magmatic zircon ε_{Hf} values (Li et al. 2015) and whole-rock ε_{Nd} values for the Xiarihamu is shown in Fig. 8b. The ranges of these isotope values for the Xiarihamu deposit are within the ranges of modern island arc basalts but significantly lower than those for the Huangshandong and Huangshanxi Fe-Ni-Cu sulfide deposits that are thought to have formed in a convergent plate margin after subduction ended (Sun et al. 2013; Zhang et al. 2011), and much higher than those for the Jinchuan deposit.

Sulfur isotopes

Sulfur isotope compositions of sulfide separates from the Xiarihamu Fe-Ni-Cu sulfide deposit are listed in Table 4. These new data, together with the data reported previously by Li et al. (2015), show that the eastern part of the deposit has more restricted $\delta^{34}\text{S}$ values than the western part (Fig. 9a). The ranges of $\delta^{34}\text{S}$ values for the eastern and western parts of the deposit are from 5.1 to 6‰ and from 2.2 to 6.8‰, respectively. The available data show no systematic variations of $\delta^{34}\text{S}$ values with rock types or ore textures within the Xiarihamu deposit. Overall, the $\delta^{34}\text{S}$ values of the Xiarihamu deposit are significantly higher than the range of typical MORB mantle ($-0.5 \pm 1‰$, Labidi et al. 2013) and slightly higher than that of the Heishan magmatic Fe-Ni-Cu sulfide deposit which formed in an arc setting (Xie et al. 2014).

Within the Xiarihamu deposit (Fig. 10a), it is interesting that the samples with the most elevated $\delta^{34}\text{S}$ values ($>5‰$) are characterized by the lowest Ni tenors (<8 wt%) whereas those with lower $\delta^{34}\text{S}$ values ($<5‰$) have much higher Ni tenors (>13 wt%). Another

Table 4 S isotopes of sulfide separates from the Xiarihamu Ni-Cu deposit

Sample	Orebody	Drill core	Depth (m)	Host rock	Sulfide texture	$\delta^{34}\text{S}$
HM-101	West	ZK15E09	259	Pl-bearing lherzolite	Disseminated	4.31
HM-102	West	ZK15E09	262	Pl-bearing lherzolite	Disseminated	4.29
HM-104	West	ZK15E09	292	Pl-bearing lherzolite	Disseminated	4.15
HM-105	West	ZK15E09	339	Pl-bearing lherzolite	Disseminated	3.78
HM-106	West	ZK15E09	343	Pl-bearing lherzolite	Disseminated	3.29
HM-107	West	ZK15E09	369	Pl-bearing lherzolite	Disseminated	4.21
HM-108	West	ZK15E09	375	Pl-bearing lherzolite	Disseminated	3.85
HM-109	West	ZK15E09	381	Pl-bearing lherzolite	Disseminated	2.81
HM-110	West	ZK15E09	390	Pl-bearing lherzolite	Disseminated	3.85
XRH-b60	West	ZK1107	152	Websterite	Disseminated	5.00
<i>XRH-b60</i>		<i>Duplicate analysis</i>				4.95
XRH-b127	West	ZK2105	431	Lherzolite	Disseminated	4.24
XRH-b26	West	ZK502	106	Lherzolite	Disseminated	3.92
XRH-b59	West	ZK1107	140	Websterite	Semi-massive	2.22
XRH-b35	West	ZK503	163	Websterite	Massive	4.99
XRH-b61	East	ZK009	115	Websterite	Disseminated	5.17
<i>XRH-b61</i>		<i>Duplicate analysis</i>				5.04
XRH-b76	East	ZK107	108	Websterite	Disseminated	5.53
XRH-b79	East	ZK107	150	Websterite	Disseminated	5.12
XRH-b91	East	ZK207	41	Websterite	Disseminated	5.95
XRH-b117	East	ZK101	182	Websterite	Semi-massive	5.21

$\delta^{34}\text{S}$ values are in per mil (‰), relative to V-CDT

interesting observation is a weak negative correlation between Rh tenors and $\delta^{34}\text{S}$ values (Fig. 10b).

Re-Os isotopes

The Re-Os isotopic compositions of the Xiarihamu deposit are listed in Table 5. The initial $^{187}\text{Os}/^{188}\text{Os}$ ratio and γOs were calculated using the zircon U-Pb age of 411.6 Ma for the host ultramafic rocks given by Li et al. (2015). The sulfide ore samples from the Xiarihamu deposit used in this study contain 0.15 to 205 ppb Re and 0.003 to 7.4 ppb Os, yielding high Re/Os ratios from 5 to 74, which are significantly higher than the fertile mantle (Shirey and Walker 1998). The samples from the eastern part of the deposit have high $\gamma\text{Os}(t)$ values varying from 1294 to 1393, which are higher than those (78–1178) for the samples from the western part of the deposit.

The data for the Xiarihamu deposit from this study show that $\gamma\text{Os}(t)$ is negatively correlated with Rh tenor (Fig. 10c) and positively correlated with $\delta^{34}\text{S}$ (Fig. 10d). Like other major magmatic Fe-Ni-Cu sulfide deposits in China such as Huangshandong and Jinchuan, the Xiarihamu deposit is characterized by elevated $\gamma\text{Os}(t)$ values that are significantly higher than that of the fertile mantle (Shirey and Walker 1998), indicating addition of crustal Os during ore formation (Fig. 11).

Modeling and discussion

Controls on PGE tenors

Good correlations between PGEs (except Pt) and S (Fig. 5) and between the individual PGEs (Fig. 6) indicate that the distribution of PGEs in the Xiarihamu deposit was primarily controlled by magmatic sulfides. The lack of correlation between Pt and S in the deposit (Fig. 5h) may be related to the nugget effect or redistribution of Pt during postmagmatic hydrothermal alteration. Studies on many magmatic sulfide deposits in the world show that the other PGE commonly occur in sulfides and semimetal sulfides, but Pt also occurs as an alloy (e.g., Dare et al. 2010; Prichard et al. 2013), which may leave it susceptible to being more mobile or more localized.

The primary controls on the concentrations of PGEs in immiscible sulfide liquids include the initial contents of these metals in the parental magma, the R factor (magma/sulfide mass ratio: Campbell and Naldrett 1979) during sulfide segregation, and fractional crystallization of monosulfide solid solution (MSS) from sulfide liquid on cooling (e.g., Li et al. 1992). All PGE partition strongly into the sulfide melt during sulfide segregation (Mungall and Brenan 2014), but the IPGEs (Os, Ir, Ru, Rh) are compatible in MSS, whereas the PPGEs (Pt, Pd) are incompatible in MSS (Li et al. 1996).

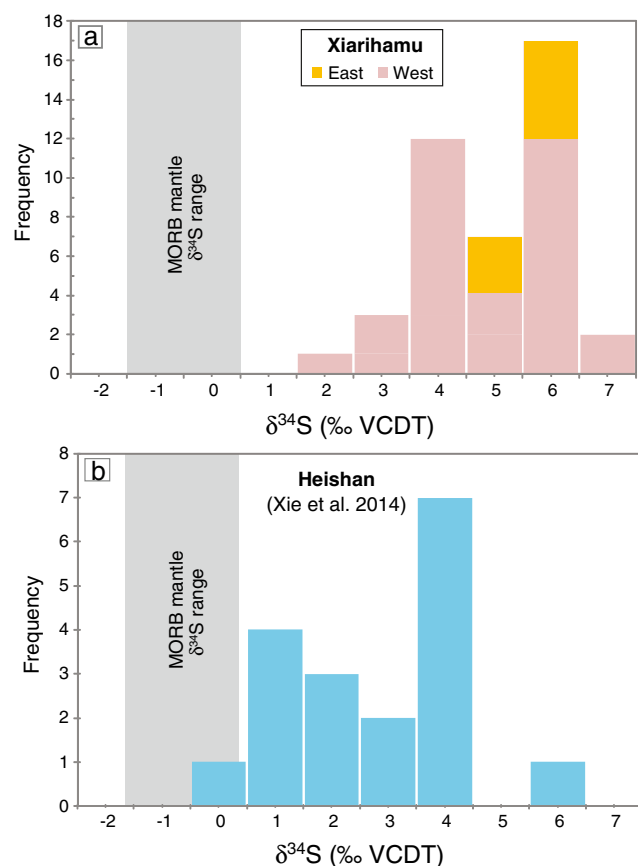


Fig. 9 Histograms of $\delta^{34}\text{S}$ values of sulfides from the Xiarihamu Ni-Cu deposit (a) and the arc-type Heishan Ni-Cu deposit (b, Xie et al. 2014). The range of $\delta^{34}\text{S}$ for mid-ocean ridge basalts (MORB) is from Labidi et al. (2013)

As shown in Fig. 5, in the Xiarihamu deposit Rh and Pd have the best correlations with S for IPGEs and PPGEs, respectively. Hence, these two elements are the best choices for modeling the primary controls on PGE fractionation in the Xiarihamu magmatic system. As shown in Fig. 6g, the samples from the Xiarihamu deposit show good correlation between Pd and Rh. This indicates that MSS fractionation is not a major control on the fractionation of PGEs in the samples. The most important controls are the initial concentrations of these metals in the parent magma prior to sulfide segregation and the R factor.

Due to their extremely high partition coefficients between sulfide liquid and magma ($>10^5$; Mungall and Brenan 2014), PGE tenors vary with their initial concentrations in magma as well as the R factor (Campbell and Barnes 1984). As a result, one of these two variables needs to be constrained independently before the other can be estimated from PGE data. Generally, it is easier to obtain reliable concentrations of Ni and Cu in magma than those of PGEs. This, together with much lower sulfide/magma partition coefficients for the base metals than PGEs, make Ni or Cu a better choice than PGEs for estimating the average R factor for a magmatic sulfide

deposit. In the case of Xiarihamu, Ni tenors were variably affected by subsolidus Fe-Ni exchange reaction between sulfide and olivine. As a result, Ni is not as good as Cu for this purpose.

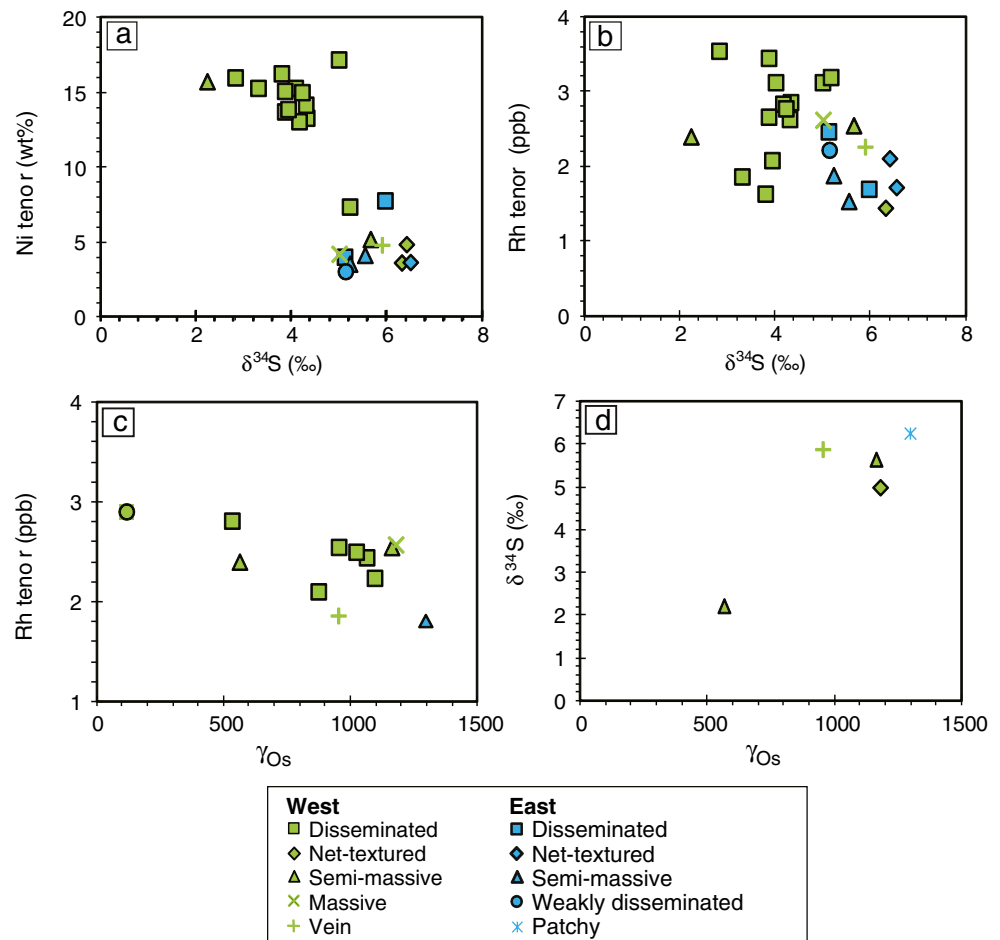
The concentrations of Cu in primitive arc basalts are up to 100 ppm (Lee et al. 2012). We have used this value to represent the initial concentration of Cu in the Xiarihamu magma prior to sulfide segregation. The average Cu tenor in the disseminated and net-textured sulfide ores of the Xiarihamu deposit is ~2 wt%. Based on this value, together with a partition coefficient of 1200 for Cu, which is within the range of experimental results for basaltic magmas (Ripley et al. 2002; Mungall and Brenan 2014), the average R factor for the disseminated and net-textured sulfide ores of the Xiarihamu deposit is estimated to be ~250. Weakly disseminated sulfide samples were excluded from the average because of large errors associated with the tenor calculation due to low S contents. Massive, semimassive sulfide ores and a vein sample were also excluded because the effect of sulfide mineral fractionation on these samples is commonly more pronounced than on disseminated and net-textured sulfide ores.

Using the average R factor estimated from Cu as a medium value for the range of R factors for the Xiarihamu deposit and assuming the partition coefficients of PGEs between sulfide melt and magma to be 10^5 , which is similar to the experimental values for basaltic magmas (Mungall and Brenan 2014), the initial contents of Rh and Pd in the parental magma are estimated to be 0.014 and 0.24 ppb, respectively (Fig. 12). These values will be reduced if higher R factors are used in the modeling. Higher D values for the PGEs have little effect on the results.

Cause of PGE depletion in parental magma

The inferred initial PGE contents in the parental magma of the Xiarihamu deposit are more than one order of magnitude lower than the values in some continental picrites associated with continental flood basalts (Lightfoot and Keays 2005; Li et al. 2016). The causes of such severe depletions include sulfide retention in the source mantle or previous sulfide segregation during magma ascent or ponding before final emplacement. The parental magma for the Xiarihamu lherzolites estimated by Li et al. (2015) contains ~10 wt% MgO and is in equilibrium with olivine containing 90 mol% Fo that is similar to mantle olivine composition. Thus, the inferred parental magma for the Xiarihamu deposit can be regarded as a primary magma, i.e., a mantle-derived magma without undergoing any differentiation such as fractional crystallization and crustal contamination. If sulfide instead of sulfate is stable in the source mantle of the Xiarihamu deposit, then the PGE depletion in the primary magma can be attributed to sulfide retention in the source mantle during partial melting, because according to the calculations by Naldrett (2011), the degree of

Fig. 10 Plots of Ni tenor vs $\delta^{34}\text{S}$ (a), Rh tenor vs $\delta^{34}\text{S}$ (b), Rh tenor vs γ_{Os} (c), and $\delta^{34}\text{S}$ vs γ_{Os} (d) for the Xiarihamu deposit



partial melting for a magma containing ~10 wt% MgO is too low to dissolve all sulfides in the mantle. Sulfide is stable in the mantle if the oxidation state is below QFM+1.5 (Jugo et al. 2005). The average oxidation state of arc basalts worldwide is QFM+1 (Kelley and Cottrell 2012). Thus, sulfide retention in the source mantle is a plausible explanation for the severe PGE depletions in the parent magma of the Xiarihamu deposit, although we cannot rule out the possibility of previous sulfide segregation due to addition of external sulfur to the Xiarihamu magma.

Source mantle characteristics and crustal contamination

The Sr-Nd isotope data shows that the parental magma of the Xiarihamu deposit was generated by mixing between a mantle-derived magma with ϵ_{Nd} close to -1.8 and $(^{87}\text{Sr}/^{86}\text{Sr})_i$ close to 0.705 and upper crustal materials (Fig. 8a). The degrees of crustal contamination are estimated to be between 5 and 30 wt%. As shown in Fig. 8a, the interception of the observed isotope array for the Xiarihamu deposit with the mantle array indicate that the source mantle for the Xiarihamu parent magma is isotopically more enriched than that of Cenozoic arc basalts worldwide and that of the

Heishan arc-type magmatic Fe-Ni-Cu sulfide deposit (Xie et al. 2014), but more depleted than that of the Jinchuan deposit which is interpreted to have formed in an intra-craton setting (Li and Ripley 2011). The significant variations in source mantle compositions among some major magmatic sulfide deposits in China are also indicated by the combination of co-magmatic zircon ϵ_{Hf} with whole-rock ϵ_{Nd} values (Fig. 8b).

The Sr-Nd-Hf isotopes are good tracers of lithophile elements while the S-Os isotopes are good tracers of highly chalcophile elements. The $\delta^{34}\text{S}$ values (Fig. 9a) and γ_{Os} values (Fig. 11) of the Xiarihamu deposit clearly indicate the addition of crustal S and Os during ore formation.

Sulfide saturation and metal upgrading

At crustal levels, sulfide saturation in basaltic magma may be induced by fractional crystallization, magma mixing or addition of crustal sulfur to the magma (Arndt et al. 2005; Keays and Lightfoot 2010; Ripley and Li 2013). In the case of Xiarihamu, fractional crystallization is the least likely cause because the inferred parental magma for the host rocks is almost as primitive as a primary magma (Li et al. 2015).

Table 5 Re-Os isotopic compositions of the Xiarihamu magmatic Ni-Cu sulfide deposit

Sample no.	Drill core	Depth (m)	Orebody	Host rock	Sulfide texture	Re (ppb)	Os (ppb)	$^{187}\text{Re}/^{188}\text{Os}$	1σ	$^{187}\text{Os}/^{188}\text{Os}$	1σ	$(^{187}\text{Os}/^{188}\text{Os})_i$	γOs
XRH-b77	ZK107	122	East	Websterite	Semi-massive	63.95	1.119	275.1882	1.1743	3.6251	0.0180	1.7315	1293.60
XRH-b108	ZK011	130	East	Websterite	Patchy	81.81	1.421	277.1974	1.1903	3.6401	0.0184	1.7328	1294.60
XRH-b75	ZK107	108	East	Websterite	Massive	77.88	1.362	275.2188	1.2328	3.6394	0.0179	1.7456	1304.96
XRH-b116	ZK101	180	East	Websterite	Massive	49.67	0.854	280.1744	1.1742	3.7826	0.0191	1.8547	1392.76
XRH-b128	ZK2105	443	West	Dunite	Disseminated	12.74	0.173	354.7004	1.9246	3.6514	0.0260	1.2107	874.44
XRH-b8	ZK905	112	West	Gabbro	Disseminated	0.15	0.003	218.4952	3.7442	1.8262	0.0397	0.3228	159.78
XRH-b56	ZK1701	298	West	Lherzolite	Disseminated	3.79	0.075	242.2038	1.8574	2.4542	0.0308	0.7877	533.93
XRH-b41	ZK1507	428	West	Lherzolite	Disseminated	1.23	0.025	235.5928	6.3494	2.4432	0.0728	0.8221	561.64
XRH-b2	ZK903	140	West	Lherzolite	Disseminated	205.39	5.258	188.0573	1.1589	2.6035	0.0130	1.3095	953.93
XRH-b48	ZK1703	301	West	Lherzolite	Disseminated	10.52	0.197	257.0537	1.3560	3.1643	0.0238	1.3956	1023.19
XRH-b104	ZK1309	231	West	Lherzolite	Disseminated	13.35	0.200	321.4413	1.6760	3.6584	0.0270	1.4466	1064.26
XRH-b31	ZK501	112	West	Orthopyroxenite	Disseminated	5.18	0.097	256.3333	1.6053	3.2484	0.0176	1.4846	1094.87
XRH-b35	ZK503	163	West	Websterite	Massive	105.78	1.967	258.8783	1.6524	3.3698	0.0168	1.5885	1178.45
XRH-b59	ZK1107	140	West	Websterite	Semi-massive	109.96	2.658	199.1558	0.8585	2.1957	0.0109	0.8253	564.24
XRH-b11	ZK905	273	West	Websterite	Semi-massive	80.62	1.555	249.6343	1.1185	3.2858	0.0163	1.5681	1162.07
XRH-b4	ZK903	214	West	Websterite	Vein	112.39	2.939	184.1037	0.8607	2.5736	0.0132	1.3068	951.75
XRH-b28	ZK502	265	West	Lherzolite	Weakly disseminated	0.89	0.116	36.9478	0.1703	0.5259	0.0051	0.2717	118.66
XRH-b15	ZK904	215	West	Websterite	Weakly disseminated	38.41	7.425	24.9034	0.1168	0.3927	0.0025	0.2213	78.13

$\lambda = 1.666 \times 10^{-11}$ /year, $T = 411.6$ Ma; the primitive initial $^{187}\text{Os}/^{188}\text{Os}$, $(^{187}\text{Os}/^{188}\text{Os})_i$, is assumed to be 0.09531 and the average $^{187}\text{Re}/^{188}\text{Os}$ of chondrites is 0.40186 (after Shirey and Walker 1998)

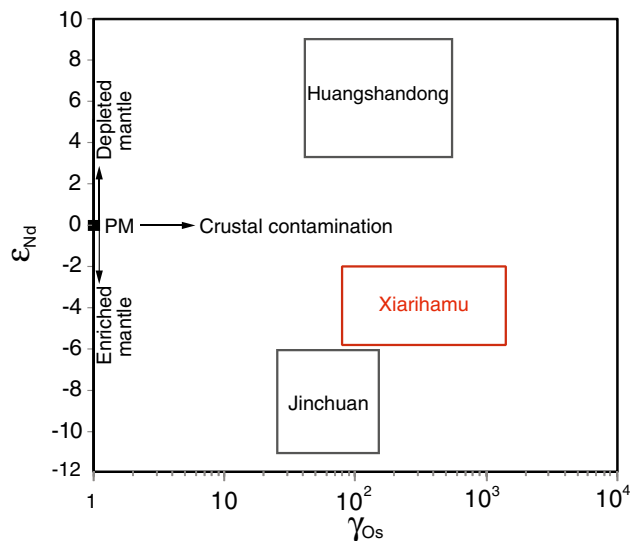


Fig. 11 Plot of ϵ_{Nd} vs γ_{Os} for the Xiarihamu deposit and two major magmatic Ni-Cu deposits in China. Data source: Xiarihamu (Jiang et al. 2015; this study), Huangshandong (Gao and Zhou 2013; Sun et al. 2013; Zhang et al. 2008; Zhou et al. 2004), Jinchuan (Li et al. 2005; Yang et al. 2005, 2008; Zhang et al. 2004)

Magma mixing is not the main reason either because this process can only produce high R factor sulfide mineralization (Li and Ripley 2005), not the low R factor sulfide mineralization in the Xiarihamu deposit (Fig. 12). Addition of crustal sulfur is the best explanation for sulfide saturation in the Xiarihamu magma. This explanation is clearly supported by the elevated $\delta^{34}\text{S}$ values (Fig. 9a) and a positive correlation between $\gamma_{\text{Os}}(\text{t})$ values and $\delta^{34}\text{S}$ values in the Xiarihamu deposit (Fig. 10d).

Upgrading of chalcophile elements in immiscible sulfide liquids by a new surge of mantle-derived magma in the Xiarihamu magmatic system is indicated by negative

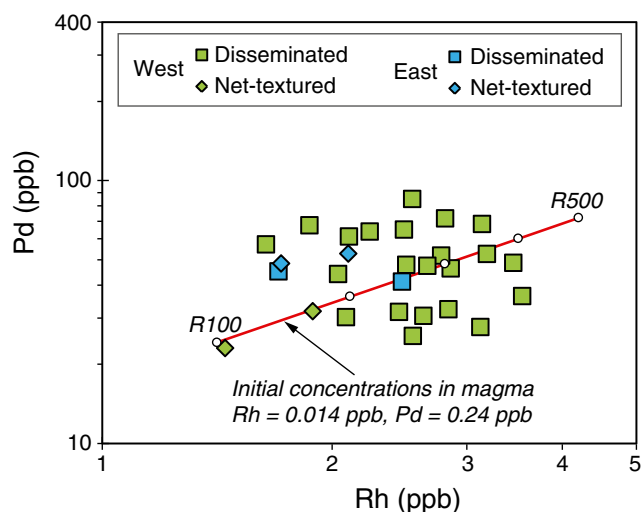


Fig. 12 Modeling of Rh-Pd tenor variations as a result of variable R factors in the Xiarihamu deposit using the equation of Campbell and Naldrett (1979). See text for explanation

correlation between Rh tenor and $\delta^{34}\text{S}$ (Fig. 10b), and between Rh tenor and γ_{Os} (Fig. 10c). Another line of evidence for this process is the inverse relationship between Ni tenors and $\delta^{34}\text{S}$ values (Fig. 10a). However, owing to subsolidus Fe-Ni exchange reaction between olivine and sulfide liquid (Li et al. 2007), which affects Ni tenors but not $\delta^{34}\text{S}$ values of sulfide ores, the relationship between Ni tenor and $\delta^{34}\text{S}$ is more complicated than that between Rh tenor and $\delta^{34}\text{S}$ (Fig. 10a, b).

Based on olivine systematics, Li et al. (2015) suggested that the Xiarihamu deposit was a dynamic magmatic system through which with multiple pulses of magma with different compositions passed through. The inverse relationship between metal tenors and $\delta^{34}\text{S}$ values described above not only supports this model but also indicates that mixing between the different pulses of magma took place at Xiarihamu.

Conclusions

New findings from this study are listed as follows.

1. The parental magma of the arc-type Xiarihamu magmatic sulfide deposit was severely depleted in PGEs, likely due to sulfide retention in the source mantle during partial melting.
2. The source mantle of the Xiarihamu Ni-Cu deposit is isotopically (Sr-Nd) more enriched than that of Cenozoic arc basalts worldwide but more depleted than that of the Jinchuan Ni-Cu deposit.
3. Addition of crustal S and Os to the Xiarihamu magma is clearly indicated by higher $\delta^{34}\text{S}$ and γ_{Os} values for the deposit than the typical mantle values.
4. Chalcophile element upgrading of previous sulfide liquids by new surge of magma at Xiarihamu is indicated based on the inverse relationship of metal tenors with $\delta^{34}\text{S}$ and γ_{Os} values.

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